

Experiment 1 — String Operations

1. Substring

```
#include <stdio.h>

int main()
{
    char str[100];
    int start, length, i;

    printf("Enter string: ");
    scanf("%s", str);

    printf("Enter starting position: ");
    scanf("%d", &start);

    printf("Enter length: ");
    scanf("%d", &length);

    printf("Substring: ");

    for(i = start; i < start + length; i++)
    {
        printf("%c", str[i]);
    }

    return 0;
}
```

OUTPUT :

Output

```
Enter string: SEJAL
Enter starting position: 2
Enter length: 3
Substring: JAL
```

2. Palindrome

```
#include <stdio.h>

int main()
{
    char str[100];
    int i, j, len = 0, flag = 0;

    printf("Enter string: ");
    scanf("%s", str);

    while(str[len] != '\0')
        len++;

    for(i = 0, j = len - 1; i < j; i++, j--)
    {
        if(str[i] != str[j])
        {
            flag = 1;
            break;
        }
    }

    if(flag == 0)
        printf("String is Palindrome");
    else
        printf("String is Not Palindrome");

    return 0;
}
```

OUTPUT:

Output

```
Enter string: 282
String is Palindrome
```

3. Compare Two Strings

```
#include <stdio.h>

int main()
{
    char str1[100], str2[100];
    int i = 0, flag = 0;

    printf("Enter first string: ");
    scanf("%s", str1);

    printf("Enter second string: ");
    scanf("%s", str2);

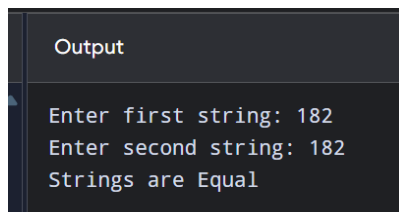
    while(str1[i] != '\0' || str2[i] != '\0')
    {
        if(str1[i] != str2[i])
        {
            flag = 1;
            break;
        }

        i++;
    }

    if(flag == 0)
        printf("Strings are Equal");
    else
        printf("Strings are Not Equal");

    return 0;
}
```

OUTPUT:

A screenshot of a terminal window with a dark background. The title bar at the top says "Output". The terminal shows the following text: "Enter first string: 182", "Enter second string: 182", and "Strings are Equal".

```
Output
Enter first string: 182
Enter second string: 182
Strings are Equal
```

4. Copy a String

```
#include <stdio.h>

int main()
{
    char str[100], copy[100];
    int i;

    printf("Enter string: ");
    scanf("%s", str);

    for(i = 0; str[i] != '\0'; i++)
    {
        copy[i] = str[i];
    }

    copy[i] = '\0';

    printf("Copied string: %s", copy);

    return 0;
}
```

OUTPUT:

Output

```
Enter string: MMCOE
Copied string: MMCOE
```

5. Reverse a String

```
#include <stdio.h>

int main()
{
    char str[100];
    int i, len = 0;

    printf("Enter string: ");
    scanf("%s", str);

    while(str[len] != '\0')
        len++;

    printf("Reverse: ");

    for(i = len - 1; i >= 0; i--)
    {
        printf("%c", str[i]);
    }

    return 0;
}
```

OUTPUT:

Output

```
Enter string: HELLO
Reverse: OLLEH
```